

How Ultrasonic Sensors Work

1. **Emission of Ultrasonic Waves:** The sensor emits a series of high-frequency sound waves (ultrasonic waves) that are above the range of human hearing (typically around 40 kHz).
2. **Reflection of Waves:** When these sound waves hit an object, they bounce back (reflect).
3. **Detection of Echo:** The sensor has a receiver that detects the reflected waves.
4. **Distance Calculation:** The time taken for the ultrasonic waves to return is measured. By knowing the speed of sound, the sensor calculates the distance to the object using the formula:

$$\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time}}{2}$$

Ultrasonic Sensor HC-SR04

Activate Windows

Key Features

01

Non-contact Measurement: Ultrasonic sensors can measure distance without needing to touch the object.

02

Wide Range: They can detect objects at various distances, typically from a few centimeters to several meters.

03

Environmental Resistance: They are not affected by light or color, making them suitable for various conditions.

```
1 // void setup() {  
2   // put your setup code here, to run once:  
3   pinMode(TrigPin,OUTPUT);  
4   pinMode(EchoPin,INPUT);  
5   Serial.begin(9600);  
6 }  
7  
8 void loop() {  
9   // put your main code here, to run repeatedly:  
10  //EMISSION OF A SOUND WAVE  
11  digitalWrite(TrigPin,LOW);  
12  delayMicroseconds(2); //2 microseconds or delay(0.002)  
13  digitalWrite(TrigPin,HIGH);  
14  delayMicroseconds(10); //10microseconds  
15  digitalWrite(TrigPin,LOW);  
16  
17  //RECEIVE OF THE REFLECTED SOUND WAVE  
18  duration=pulseIn(EchoPin,HIGH); //measure the duration when the echo pin is HIGH  
19  
20  |  
21  Serial.print("Distance :");  
22  Serial.print(distance);  
23  }
```

```
1 int TrigPin=11; //transmitted sound wave from the sensor
2 int EchoPin=12; //received sound wave to the sensor
3
4 long duration, inches;
5
6 void setup() {
7   // put your setup code here, to run once:
8   pinMode(TrigPin,OUTPUT);
9   pinMode(EchoPin,INPUT);
10  Serial.begin(9600);
11 }
12
13 void loop() {
14   // put your main code here, to run repeatedly:
15   //EMISSION OF A SOUND WAVE
16   digitalWrite(TrigPin,LOW);
17   delay(0.002); //2 microseconds
18   digitalWrite(TrigPin,HIGH);
19   delay(0.001);
20   digitalWrite(TrigPin,LOW);
21
22   //RECEIVE OF THE REFLECTED SOUND WAVE
```

Ultrasonic Sensors Specifications



Operating Voltage	5V DC
Operating Current	15mA
Operating Frequency	40KHz
Min Range	2cm / 1 inch
Max Range	400cm / 13 feet
Accuracy	3mm
Measuring Angle	<15°
Dimension	45 x 20 x 15mm

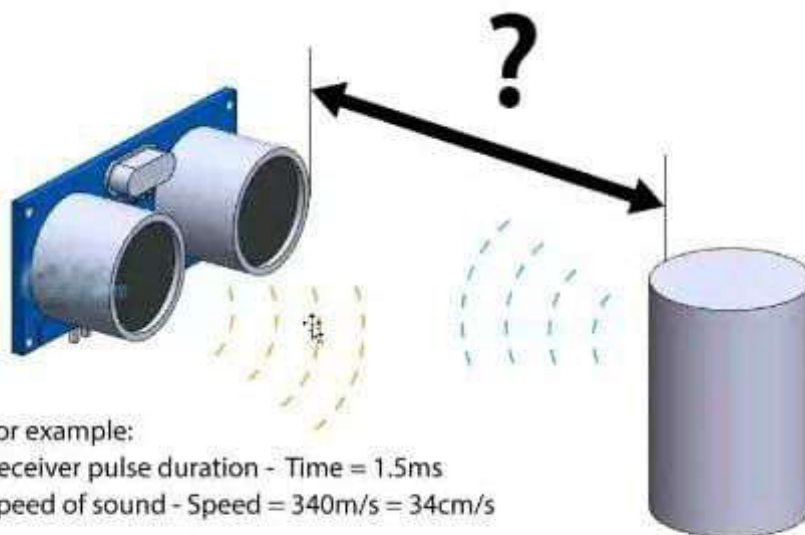
sketch_aug5b.ino

```
4
5 void setup() {
6   // put your setup code here, to run once:
7   pinMode(TrigPin,OUTPUT);
8   pinMode(EchoPin,INPUT);
9   Serial.begin(9600);
10 }
11
12 void loop() {
13   // put your main code here, to run repeatedly:
14   //EMISSION OF A SOUND WAVE
15   digitalWrite(TrigPin,LOW);
16   delay(0.002); //2 microseconds
17   digitalWrite(TrigPin,HIGH);
18   delay(0.001);
19   digitalWrite(TrigPin,LOW);
20
21   //RECEIVE OF THE REFLECTED SOUND WAVE
22   duration=pulseIn(EchoPin,HIGH); //measure the duration when the echo pin is HIGH
23   distance; //distance = speed of sound x duration /2 (we divide 2 for the round trip of the wave)
24
25 }
```

Activate Windows
Go to Settings to activate Windows.

Ln 23, Col 96: ;
Arduino Uno

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For example:

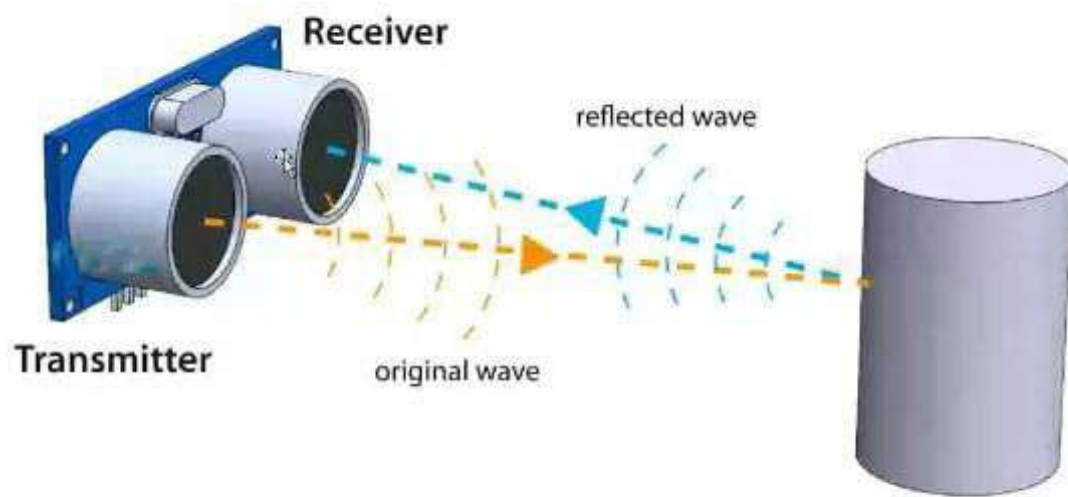
Receiver pulse duration - Time = 1.5ms

Speed of sound - Speed = 340m/s = 34cm/s

Distance = (Speed x Time) / 2

Distance = (34cm/ms x 1.5ms) / 2 = 25.5cm

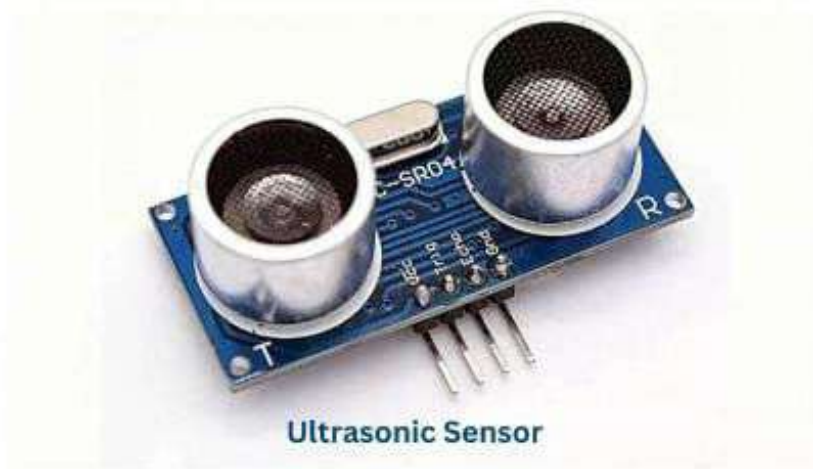
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```
1 int TrigPin=11; //transmitted sound wave from the sensor
2 int EchoPin=12; //received sound wave to the sensor
3 int duration, distance;
4
5 void setup() {
6   // put your setup code here, to run once:
7   pinMode(TrigPin,OUTPUT);
8   pinMode(EchoPin,INPUT);
9   Serial.begin(9600);
10 }
11
12 void loop() {
13   // put your main code here, to run repeatedly:
14
15 }
16
```

Ultrasonic sensor

An ultrasonic sensor is an electronic device that uses ultrasonic waves to measure distance, detect objects, or sense the presence of nearby items. These sensors are widely used in various applications due to their accuracy and reliability.



```
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5 void setup() {
6   // put your setup code here, to run once:
7   pinMode(TrigPin,OUTPUT);
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9   Serial.begin(9600);
10 }
11
12 void loop() {
13   // put your main code here, to run repeatedly:
14   //EMISSION OF A SOUND WAVE
15   digitalWrite(TrigPin,LOW);
16   delay(0.002); //2 microseconds
17   digitalWrite(TrigPin,HIGH);
18   delay(0.001);
19   digitalWrite(TrigPin,LOW);
20
21   //RECEIVE OF THE REFLECTED SOUND WAVE
22   duration=pulseIn(EchoPin,HIGH); //measure the duration when the echo pin is HIGH
23   distance = duration*0.034/2; //distance = speed of sound x duration / 2 (we divide 2 for the round trip of the
24   //sound speed = 340m/s or 34cm/s or 0.034cm/ms
25 }
```

Applications

Ultrasonic sensor Applications



Obstacle Detection

Used in robotics for navigation and collision avoidance.



Level Sensing

Used in tanks and silos to measure liquid or solid levels.



Parking Assistance

Used in vehicles to help with parking by detecting obstacles.